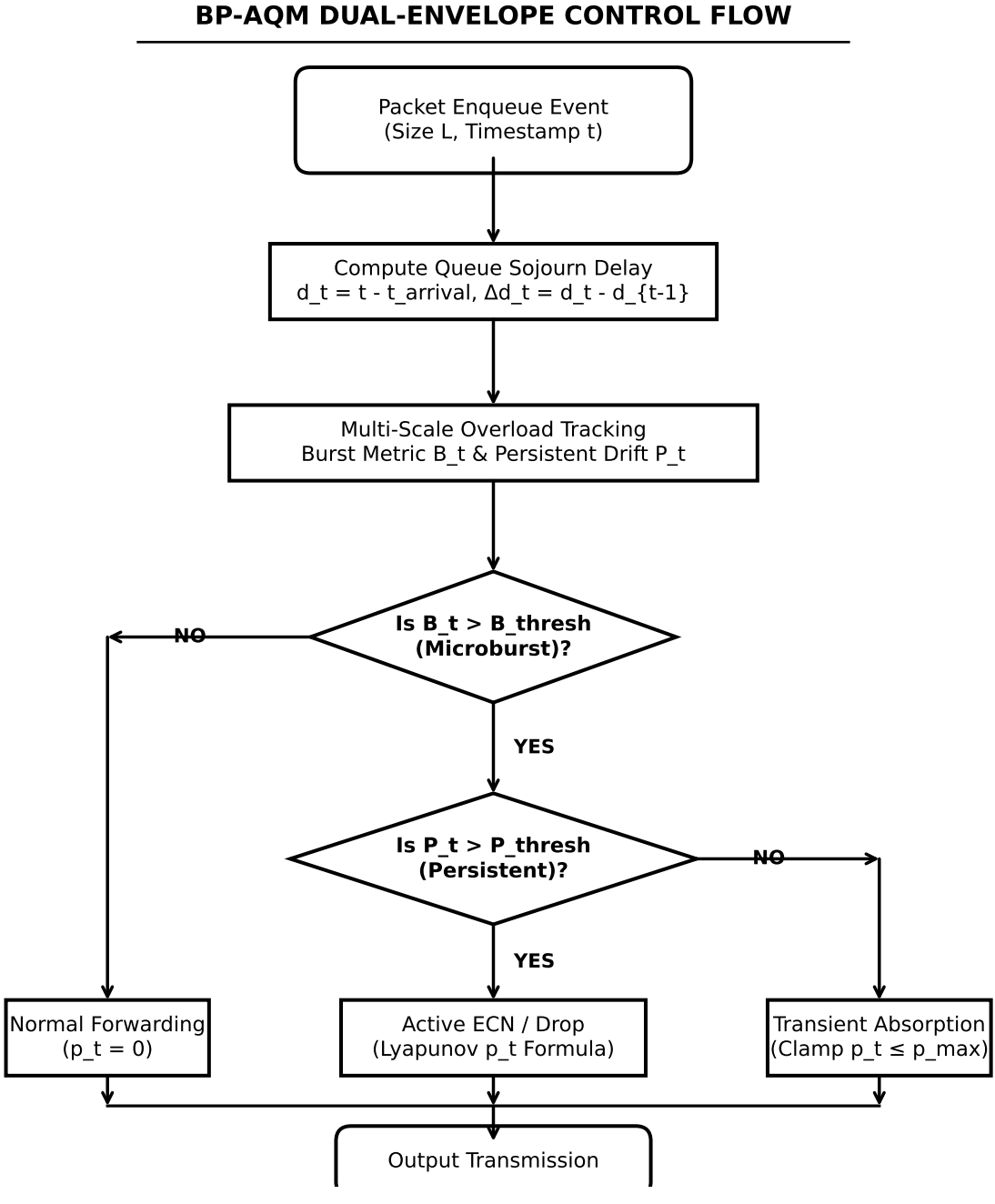
BP-AQM DUAL-ENVELOPE CONTROL FLOW
Packet Enqueue Event
(Size L, Timestamp t)
Compute Queue Sojourn Delay
d_t = t - t_arrival, Δd_t = d_t - d_{t-1}
Multi-Scale Overload Tracking
Burst Metric B_t & Persistent Drift P_t
Is B_t > B_thresh
(Microburst)?
NO
YES
Is P_t > P_thresh
(Persistent)?
NO
YES
Normal Forwarding
(p_t = 0)
Active ECN / Drop
(Lyapunov p_t Formula)
Transient Absorption
(Clamp p_t ≤ p_max)
Output Transmission